SILTHAUL · DESIGN DECISION RECORD

SiltHaul second box on the return leg and flat-pack frame

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/silthaul		
Source	docs/decisions/0003-amish-requirement-decisions.md @ 9347c2b		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Amish's requirement decisions 2A (R1) and 7A (R9) carried into the constructable design

0003: Second box on the return leg and flat-pack frame

- **Date:** 2026-10-03
- **Status:** accepted

Context

At TRL 3 (SLH-CAL-001 v0.2) SiltHaul missed R1, output, by a wide margin (0.32 m³ an hour, a ratio of 0.36 to a bucket crew against a target of 2), and R9, portability, was at risk because the welded frame, 890 x 663 x 814 mm, needed an estate car or small pickup. Both fixes change what the product is, so they were put to Amish as requirement decisions. On 2026-10-03 Amish chose option A on every decision put to him: "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A". For SiltHaul:

- **2A (R1).** Add a second box on the return run so the rope loop carries mud both ways; restate R1 as "no lifting or carrying of mud; output at least 0.5 times the same crew with shovels and buckets".
- **7A (R9).** Redesign the frame as a bolted flat-pack frame that fits a small car boot, keeping the 25 kg heaviest-part limit.

Options considered

For 2A the open question was how two boxes fit through a 0.7 m door (R4) with their rope lines 300 mm apart, the spacing the drum and ramp already set.

Option	Result	Chosen
Two 446 mm boxes, lines moved about 540 mm apart	Ramp and door about 1.1 m wide; breaks R4	No
Two narrow boxes on the existing 300 mm lines	210 mm inside, 845 mm long, 280 mm sides: 40.3 L each; they pass 42 mm apart and cross the ramp 26 mm inside its cheeks	Yes

For 7A the frame could come apart into two welded side frames joined by bolted cross members, or into loose tubes (rails, posts and braces each bolted). The side frames need only twelve bolts and keep every welded joint that carries the drum and crank bearings; loose tubes would fit a boot with the seats up but need about forty bolts and would push setup well past 20 min (R8).

Decision

Table 1. Changes made.

#	Part	Change	Result (SLH-CAL-001 v0.3)
1	Scraper boxes	Two identical boxes, 210 mm wide inside, 845 mm to the lip, 280 mm sides, back sloping 110 mm; box 1 on the pull rope, box 2 on the return rope, both open toward the door	40.3 L each at a 245 mm fill (R3); 16.1 kg each
2	Ropes	Pull rope 17 m to box 1; return rope 17 m to box 2; a tail rope joins the two boxes' rear bridles round the tail sheaves, chosen for the room from 12.5, 9 and 6 m	62 m of rope in all
3	Operation	Each stroke hauls one full box out while the other goes back empty; loaders fill one box while the crank crew dump the other	3.54 min a stroke; 0.68 m³/h, 0.51 with rest; ratio 0.57 (R1 met on paper)
4	Ramp	Centred midway between the two box lines	Both boxes 26 mm inside the cheeks (R4)
5	R1	Restated in Amish's words above	Met on paper
6	Capstan frame	Two welded side frames (rail, posts, braces, pads, top plate, stake tubes) and four bolted cross members (front and rear cross rails, top tie, anchor bar with eye), each with 8 mm end plates; eight M10 and four M12 bolts	Side frames 9.2 and 9.6 kg; cross members 8.0 kg in all
7	Stake tubes	Moved to 280 mm in front of and 330 mm behind the drum axis to clear the bolts	Stakes unchanged
8	R9	Requirement unchanged; small car boot taken as a small hatchback with the rear seats folded (1,250 x 950 x 500 mm)	Heaviest lift 23.1 kg; kit about 454 L, 77 % of the boot (R9 met on paper)

Consequences

- `cad/src/model.py` adds box 2, its bridles, the tail rope and the bolted frame, with checks that the boxes pass with at least 25 mm between them, that both cross the ramp at least 15 mm inside the cheeks, and that every cross member and bolt touches the side frames. STEP and STL, SLH-DWG-001 and 002 (Rev P3), the concept media and the build plan pictures are regenerated.
- The working pull stays 1,000 N: the narrower lip cuts less mud, so a full box out and an empty box back need about 970 N. Handle force, rope, chain, shear pin and anchors are unchanged.
- Setup rises to about 20 min with three people, at the R8 limit.
- Value-engineering target: USD 2,000. Estimated cost of the constructable design: USD 1,055.60 (USD 944.40 under the target), up USD 168.10.
- Boxes pass 42 mm apart in the room: loaders keep hands and feet out from between them while the cranks turn.
- New open decision for Amish: whether "fits a small car boot" may assume the rear seats are folded (design decisions register).